Income's Influence on Eating Out

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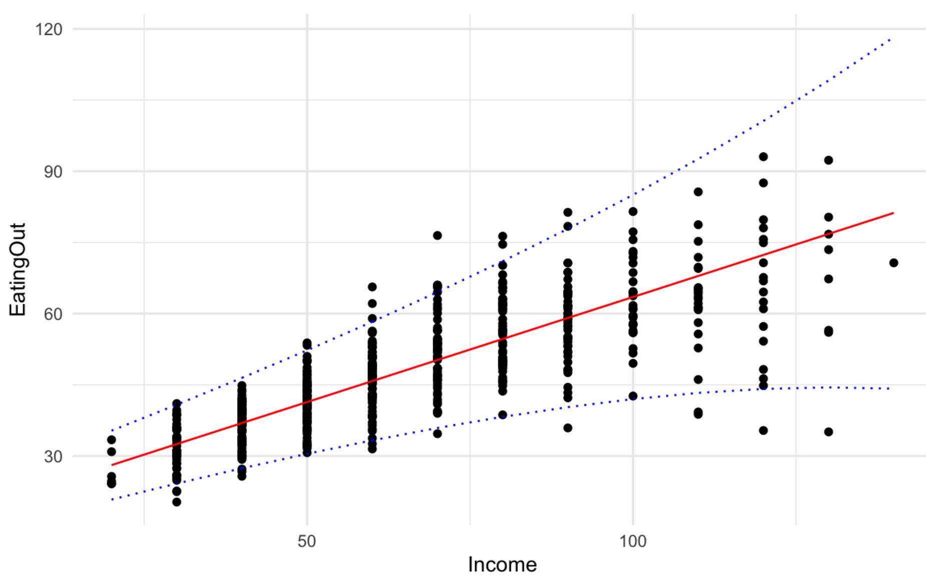
We were hired by the National Restaurant Association to investigate how customers' "eating out" behavior is influenced by their income level. This information is helpful to determine if restaurant ownership is a viable career choice in developing economic areas. We were also tasked with determining to the best of our ability how demand for "quality" and "convenience" changes as income changes (i.e. contributing to the higher variability as income increases). The data was gathered from the Food Demand Survey (FooDS) on 523 households' income and "eating out" expenses.

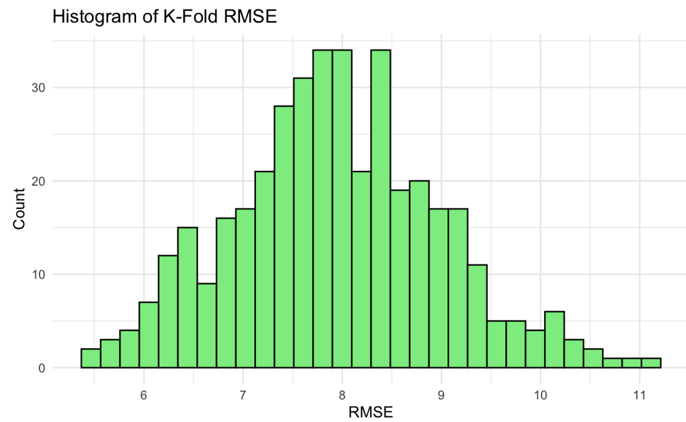
Main Conclusions

Primary Conclusions

We discovered a positive linear relationship between income and money spent eating out. There was heteroskedasticity (unequal variance) in a standard linear model, so we fit a Heteroskedastic Linear Regression model instead. In the graph below, you can see the HLR model with a 95% prediction interval, which follows the increasing variance in "eating out" expenses as Income increases.



Our model has an R^2 value of 0.6258, indicating that our model explains about 63% of the variability in the response (how much money people spend eating out). Using K-fold cross-validation, we got an average RMSE of 7.94. Because this error is smaller than the response's standard deviation (12.90), the model's predictions are meaningfully closer to the observed values than what we would expect from a naive or random baseline.



Knowing that our model does a good job at explaining the food data, we can use our model to analyze the relationship between eating out behavior and increase in income. We are 95% confident that the weekly increase in eating out per $1000 increase in income (β_1) is likely between $0.42 and $0.47.

Hypothesis Test

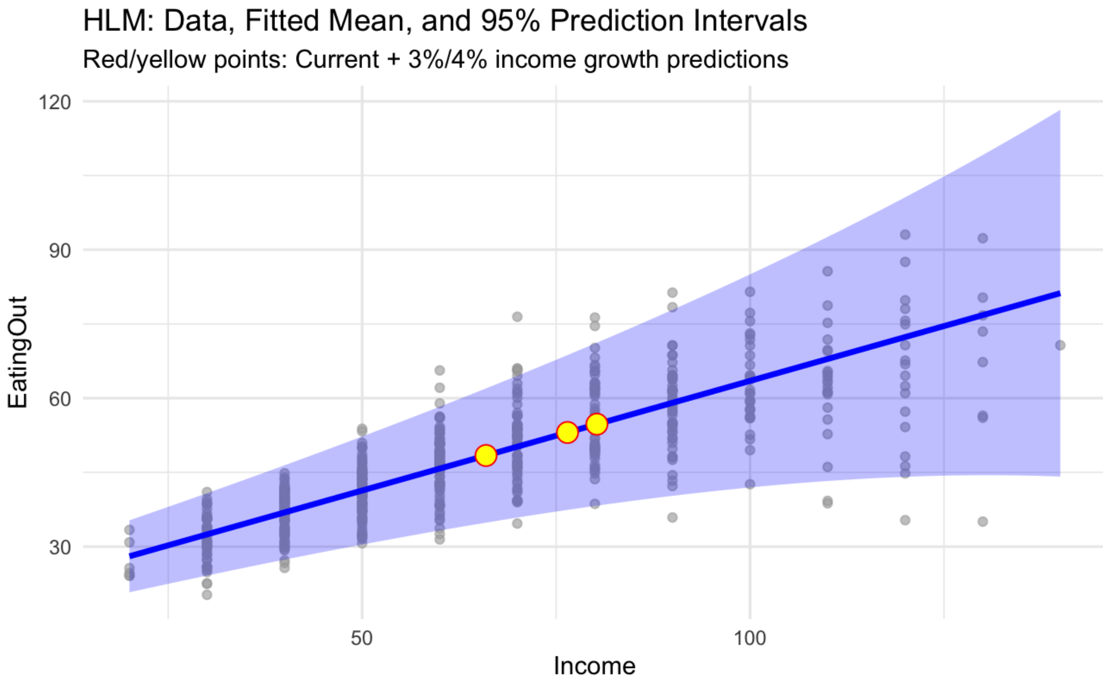
Hypotheses:

H_0: weekly increase in eating out per $1000 increase in income (β_1) $\leq$ $0.50

H_a: weekly increase in eating out per $1000 increase in income (β_1) > $0.50

Using our Heteroskedastic Linear Regression model, we performed a hypothesis test with our estimated weekly increase in eating out (per $1000 increase in income) at 0.443, with a standard error of 0.014, and got a p-value of 0.9999772. With such a large p-value, we fail to reject the null hypothesis (H_0) that β_1 is less than a $0.50 weekly increase, and conclude that the weekly change in spending for "eating out" is less than $0.50 for each $1000 increase in income and is therefore unhealthy. Additionally a 95% confidence interval shows that the weekly increase in eating out per $1000 increase in income (β_1) is likely between 0.4156425 and 0.4704193, which does not include 0.5, and further confirms that the increase is not enough (less than 0.5) and is unhealthy.

5 Year Predictions

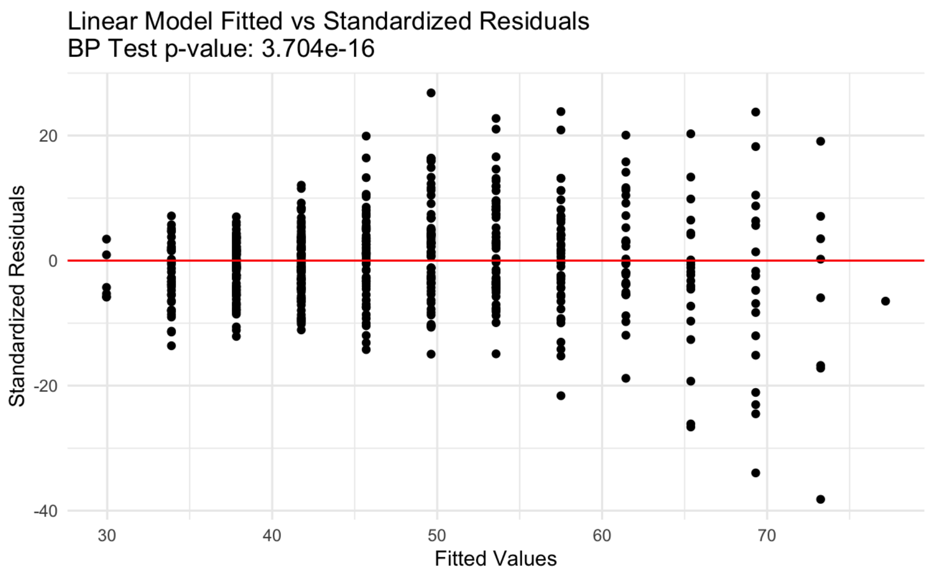


The advisers anticipate that household incomes will grow by approximately 3–4% per year. Using our heteroskedastic linear regression model, we forecast weekly EatingOut spending under different income growth scenarios. If incomes remained constant over the next five years, the model predicts an average spending of $48.41 per week, with a variance of $6.92. We predict that a household at this income level will spend between $34.86 and $61.97 per week eating out.With a 3% annual increase in income, predicted average spending rises to $53.07 per week, with variance increasing to $7.98. At a 95% prediction interval, we predict that an individual household at this income level would likely spend between $37.43 and $68.71 eating out each week.With a 4% annual income increase, the predicted average is $54.75 per week, and standard deviation rises to $8.40. With the 4% annual growth, we predict that a household would spend between $38.29 and $71.21 eating out per week. These projections indicate that as incomes increase, average spending on EatingOut will rise, but spending will also become more variable across households, particularly at higher incomes. The graph shown above illustrates these predictions.

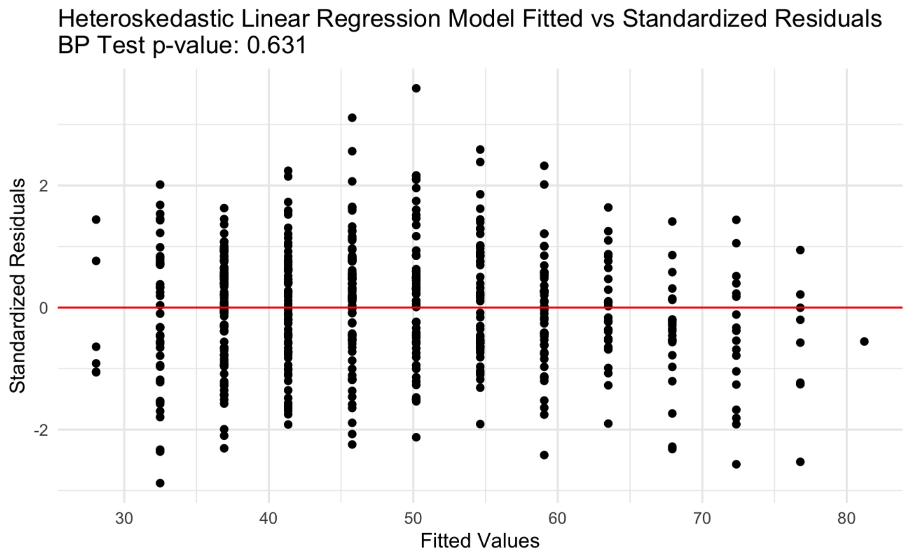
Modeling Details

Justify Why Standard Regression Model doesn't work

A standard regression model won't work for this situation because it assumes linearity, independence, normality, and equal variance (LINE). However, the data for the eating out behavior and income levels do not meet all these assumptions since equal variance is not met.



While checking LINE assumptions for a basic linear model, you can clearly see unequal variance in the residuals, verified by a BP test with a really small p-value. Unequal variance prevents you from creating any sort of reliable prediction intervals, and since our standard errors are all messed up, we can't analyze a healthy economy such as the hypothesis test we did above, essentially making the standard regression model useless. Thus we need our new model. As shown in the Fitted vs Residuals plot below, the heteroskedastic linear regression model, fixes our equal variance problem, allowing us to actually make accurate predictions.



Describe New Model

Our heteroskedastic linear regression model can be conceptualized by the following:

$$\text{EatingOut} \sim N(\mu(\text{Income}), \sigma^2(\text{Income}))$$

$$\mu(\text{Income}) = \beta_0+\beta_1*\text{Income}$$

$$\log\{\sigma(\text{Income})\} = \theta_0 +\theta_1* \text{Income}$$

Where β_0 is the intercept and β_1 is the change in weekly EatingOut spending for each $1000 increase in income.

θ_0 : This is the baseline log–standard deviation of EatingOut when Income is 0 (on whatever scale Income is measured). After exponentiating, $\exp(\theta_0)$ is the baseline standard deviation of EatingOut at Income = 0.

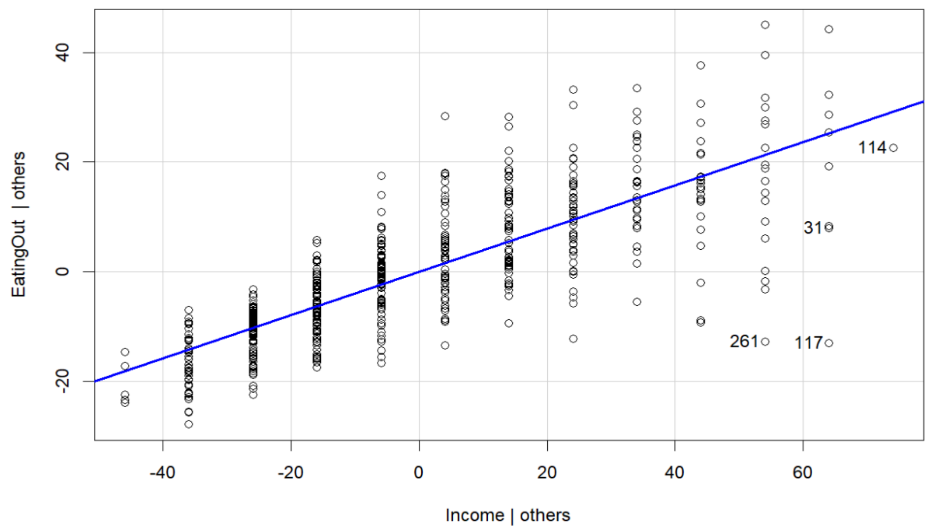
θ_1: This describes how the log–standard deviation changes with Income. A one-unit increase in Income changes $\log\sigma$ by θ_1, so it multiplies the standard deviation by $\exp(\theta_1)$.

When $\theta_1 > 0$: variability in EatingOut increases as Income increases (the spread of restaurant spending gets larger for higher-income households).

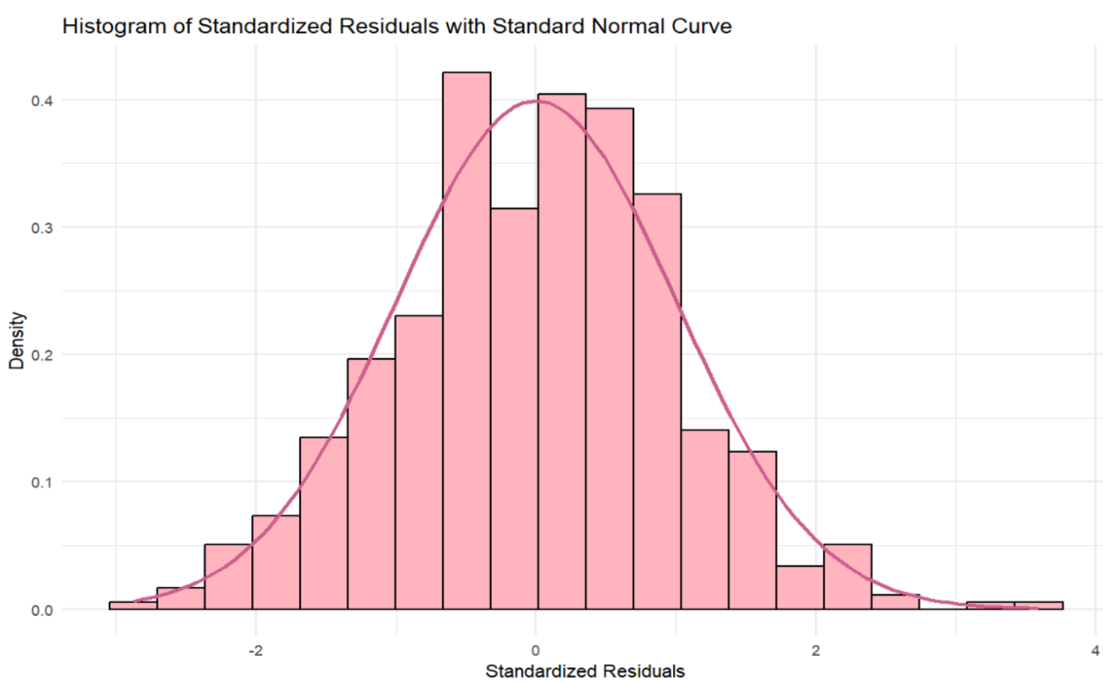
When $\theta_1 < 0$: variability in EatingOut decreases as Income increases (restaurant spending becomes more tightly clustered among higher-income households).

Validate New Model

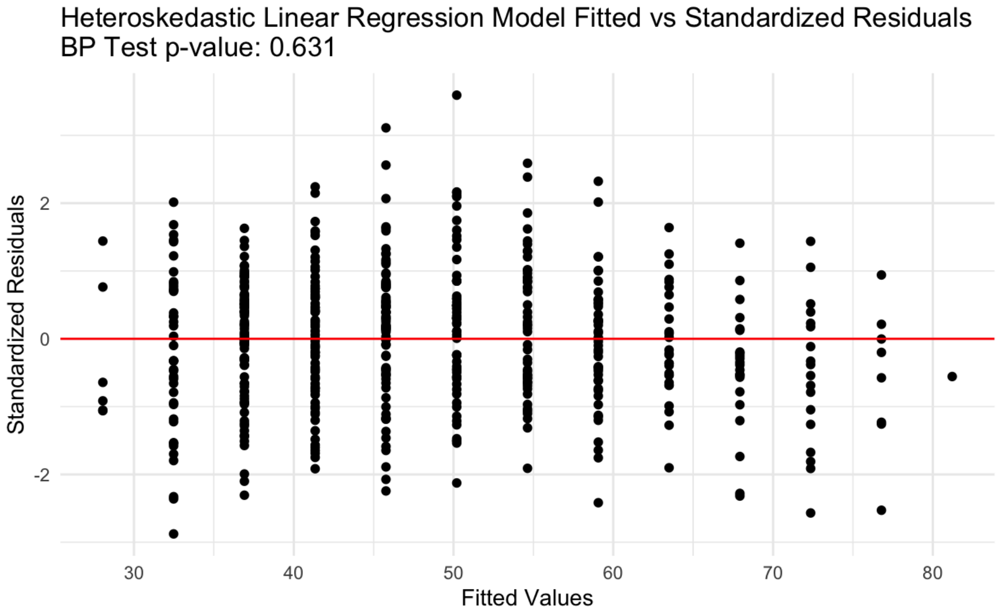
By looking at the added variable plot, one can notice a trend that as income increases, eating out increases. This model is clearly linear.



By looking at a histogram of the residuals, we can see that the normality of residuals is met for our model. In addition to looking at the histogram, the KS test revealed a high p-value of 0.8098 indicating a strong normal distribution.



As discussed before, the Equal Variance assumption is met because we got a bp test p-value of 0.631 which is greater than 0.05, and there is no obvious pattern in the fitted vs residuals plot.



Code:

````
---
title: "Food Expenditures Consulting Project"
format: html
---

```{r}
#| output: False
library(vroom)
library(tidyverse)
library(gamlss)
```

```{r}
food <- vroom("FoodExpenses.txt", col_types = "dd")
head(food)
```

## Check Line Assumptions
Linearity
```{r}
food_lm <- lm(EatingOut ~ Income, data = food)
car::avPlot(food_lm, "Income")
```

Equal Variance
```{r}
bp <- lmtest::bptest(food_lm)

bp_pval <- formatC(bp$p.value, format = "e", digits = 3)

ggplot(data.frame(fitted = fitted(food_lm), r =
residuals(food_lm)),
       aes(x = fitted, y = r)) +
  geom_point() +
  geom_hline(yintercept = 0, color = "red") +
````
````

````
  labs(title = paste0("Linear Model Fitted vs Standardized
Residuals\n",
                    "BP Test p-value: ", bp_pval),
       x = "Fitted Values",
       y = "Standardized Residuals") +
  theme_minimal()

```


GAMLSS
```{r}
gamlss_food <- gamlss(
  EatingOut ~ Income,
  sigma.formula = ~ Income,
  data = food
)
```

```{r}
pred <- predict(gamlss_food, what = "mu", type = "response")  #
fitted mean[web:26][web:27]
sigma_hat <- predict(gamlss_food, what = "sigma", type =
"response")  # fitted sigma[web:26][web:27]

# for (approximately) Normal family, 95% prediction interval:
lwr <- pred + qNO(0.025, mu = 0, sigma = sigma_hat)
upr <- pred + qNO(0.975, mu = 0, sigma = sigma_hat)

plotdat <- food |>
  mutate(
    fit = pred,
    lwr = lwr,
    upr = upr
  )

ggplot(plotdat, aes(x = Income, y = EatingOut)) +
  geom_point() +
  geom_line(aes(y = fit), color = "red") +
  geom_line(aes(y = lwr), linetype = "dotted", color = "blue") +
````

````
  geom_line(aes(y = upr), linetype = "dotted", color = "blue") +
  theme_minimal()
```

```{r}
# normality
# standardize resids

standardized_resids <- residuals(gamlss_food, what="z-scores")

# histogram of standard resids
ggplot(data.frame(resid = standardized_resids), aes(x = resid))
+
  geom_histogram(aes(y =  after_stat(density)), bins = 20, color
= "black", fill = "lightpink") +
  stat_function(fun = dnorm, args = list(mean = 0, sd = 1),
color = "hotpink3", size = 1) +
  labs(
    x = "Standardized Residuals",
    y = "Density",
    title = "Histogram of Standardized Residuals with Standard
Normal Curve"
  ) +
  theme_minimal()
```

```{r}
ks.test(standardized_resids, "pnorm", mean = 0, sd = 1)
```

```{r}
fitted_vals <- fitted(gamlss_food)

bp_test <- lmtest::bptest(
  standardized_resids ~ Income,
  data = food
)

bp_pval <- round(bp_test$p.value, 3)
````

````
ggplot(data.frame(fitted = fitted_vals, r =
standardized_resids),
       aes(x = fitted, y = r)) +
  geom_point() +
  geom_hline(yintercept = 0, color = "red") +
  labs(title = paste0("Heteroskedastic Linear Regression Model
Fitted vs Standardized Residuals\n",
                    "BP Test p-value: ", bp_pval),
       x = "Fitted Values",
       y = "Standardized Residuals") +
  theme_minimal()
```

## Cross-validation

```{r}
#| output: False
set.seed(123)

B <- 50
K <- 8
rmse_all <- c()

for(b in 1:B){
  folds <- sample(rep(1:K, length.out = nrow(food)))
  rmse_vals <- numeric(K)

  for(k in 1:K){
    train <- food[folds != k, ]
    valid <- food[folds == k, ]

    fit_k <- gamlss(
      EatingOut ~ Income,
      sigma.formula = ~ Income,
      data = train
    )
    preds <- predict(fit_k,
                     data = train,
````

````
                     newdata = valid)
    rmse_vals[k] <- sqrt(mean((valid$EatingOut - preds)^2))
  }
  rmse_all <- c(rmse_all, rmse_vals)
}
```

```{r}
ggplot(data.frame(RMSE = rmse_all), aes(x = RMSE)) +
  geom_histogram(color = "black",
                 fill = "lightgreen",
                 bins = 30) +
  labs(title = "Histogram of K-Fold RMSE",
       x = "RMSE",
       y = "Count")
```

```{r}
mean(rmse_all)
sd(food$EatingOut)
```

```{r}
R2 <- cor(food$EatingOut, fitted(gamlss_food))^2
R2
```

## Hypothesis Test
```{r}
# H0: beta_income < 0.5   vs   Ha: beta_income >= 0.5

# Get the coefficients and their covariance matrix for mu
coef_mu <- coef(gamlss_food, what = "mu")          # named
vector of estimates
vcov_mu <- vcov(gamlss_food, what = "mu")          #
variance-covariance matrix

# Extract estimate and its standard error for Income
b1_hat <- coef_mu["Income"]
````

````
se_b1  <- sqrt(vcov_mu["Income", "Income"])

# null
beta0  <- 0.5

# wald stat and p-val
z_stat <- (b1_hat - beta0) / se_b1
p_val  <- 1 - pnorm(z_stat)   # one-sided, Ha: beta >= 0.5

cat("Beta_1 hat:", b1_hat, "\n")
cat("Standerd Error of beta_1 hat:", se_b1, "\n")
cat("Z-stat:", z_stat, "\n")
cat("P-val", p_val, "\n")
```

```{r}
confint(gamlss_food, what = "mu")
```

## predictions for 3-4% increase in income

```{r}
mean_income <- mean(food$Income)

income_future <- data.frame(
  scenario = c("Current", "5yr_3pct", "5yr_4pct"),
  Income = c(
    mean_income,
    mean_income*(1.03^5),
    mean_income*(1.04^5)
  )
)

income_future

pred_data <- data.frame(
  Income = income_future$Income
)
````

````
income_future$mu_pred <- predict(
  gamlss_food,
  newdata = pred_data,
  what = "mu",
  type = "response"
)
income_future$mu_pred

income_future$sigma_pred <- predict(
  gamlss_food,
  newdata = pred_data,
  what = "sigma",
  type = "response"
)
income_future$sigma_pred
```

```{r}
# 95% prediction intervals
income_future$lower_PI <- income_future$mu_pred - 1.96 *
income_future$sigma_pred
income_future$upper_PI <- income_future$mu_pred + 1.96 *
income_future$sigma_pred

income_future
```

```{r}
# Predictions
income_seq <- seq(min(food$Income), max(food$Income), length.out
= 100)
pred_df <- data.frame(Income = income_seq)

mu_pred <- predict(gamlss_food, newdata = pred_df, what = "mu",
type = "response")
sigma_pred <- predict(gamlss_food, newdata = pred_df, what =
"sigma", type = "response")
````

````
# 95% prediction interval: mu ± 1.96 * sigma (standard for
GAMLSS plots)
pred_plot_df <- data.frame(
  Income = income_seq,
  mu = mu_pred,
  lower_pi = mu_pred - 1.96 * sigma_pred,
  upper_pi = mu_pred + 1.96 * sigma_pred
)
```

```{r}
# 95% "confidence" (actually predictive) intervals assuming
Normal(mu, sigma)
income_future <- income_future |>
  mutate(
    lower_95 = mu_pred - 1.96 * sigma_pred,
    upper_95 = mu_pred + 1.96 * sigma_pred
  )

income_future
```

```{r}
ggplot(food, aes(x = Income, y = EatingOut)) +
  geom_point(alpha = 0.6, color = "darkgray") +
  geom_line(data = pred_plot_df, aes(x = Income, y = mu),
            color = "blue", linewidth = 1.2) +
  geom_ribbon(data = pred_plot_df,
              aes(x = Income, ymin = lower_pi, ymax = upper_pi),
              alpha = 0.25, fill = "blue", inherit.aes = FALSE)
+
  geom_point(data = income_future,
             aes(x = Income, y = mu_pred),
             color = "red", size = 4, shape = 21, fill =
"yellow",
             inherit.aes = FALSE) +
  labs(
    title = "GAMLSS: Data, Fitted Mean, and 95% Prediction
Intervals",
````

```
    subtitle = "Red/yellow points: Current + 3%/4% income growth predictions",
    x = "Income", y = "EatingOut"
  ) +
  theme_minimal()
```